

**TREATMENT AND EMERGENCY PROTOCOL FOR
GLYCOGEN SYTHETASE DEFICIENCY (a Glycogen Storage Disease, GSD)**

RE:
DOB:
MR#

This patient has Glycogen Synthetase Deficiency (a type of GSD)
and is at risk for serious complications from this disorder.

Definition: Glycogen Synthetase Deficiency is a rare inherited problem of glucose metabolism. Affected persons cannot synthesize glycogen therefore have little glucose reserve. Also, they cannot metabolize excess glucose so they can become hyperglycemic, acidotic (lactic acidosis) and ketotic.

Sign of Metabolic Decompensation: Either hypoglycemia with ketosis OR hyperglycemia with lactic acidosis.

Treatments:

1. Prevention

- a. Avoidance of fasting for long periods ("Long periods" are relative and depend on age).
- b. Frequent feeding of carbohydrates in a balanced, low fat and low sugar diet.
- c. An evening/bedtime meal and supplementation with uncooked cornstarch at bedtime may be prescribed.

2. Treatment of Hypoglycemia (blood glucose <60 mg/dl)

- a. D10/W 100ml push
- b. Recheck blood glucose
- c. Continue until blood glucose is >100 mg/dl
- d. Avoid hyperglycemia (glucose >120 mg/dl)
- e. For longer maintenance use D10 ½ NS with 20 meq KCl at maintenance (1500 ml/m²/24 hours or as calculated for young children)
- f. Check blood glucose every 15 minutes until stable.
- g. After IV is stopped, continue to check glucose and observe oral intake of carbohydrate and only dismiss when blood glucose is stable in the range of 80-110 mg/dl



Notify the metabolic specialist immediately (after business hours this number allows you to speak with the geneticist on call:

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